Mingyuan Zhang

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Ph.D. candidate in Computer Engineering at Northeastern University, focused on **efficient foundation models** and **embodied AI**. Research spans efficient post-training, sparse optimization, mixture-of-experts upcycling, and agent systems, with experience in world modeling and robot-policy learning at Amazon and image restoration at Socure.

EDUCATION

***Northeastern University* (NEU)** Boston, MA
- **Ph.D.** candidate in **Computer Engineering** at SmileLab — Sep. 2023 – May 2028 (expected)
- Advisor: Prof. Yun Raymond Fu
- **Research Interest:** Foundation Model | Efficiency | Computer Vision | Embodied AI | World Model | Agents | Generative Model

***Northeastern University* (NEU)** Boston, MA
- **Master of Science** in **Electrical and Computer Engineering** — Sep. 2021 – Dec. 2022
- **Concentration:** Computer Vision, Machine Learning and Algorithm

***Xidian University* (XDU)** Xi'an, China
- **Bachelor of Engineering** in **Telecommunications Engineering** — Sep. 2017 – Jun. 2021
- **Honor of Excellent Student Leader**, 2020

EXPERIENCE

Northeastern University | **SmileLab** | *Research Assistant* | Boston, MA — *Sep. 2023 – Present*

Advisor: Prof. Yun Raymond Fu

Efficient Foundation Model Post-Training [MFT] [Rethinking] [MASKerade]
- Proposed structural post-training methods that adapt frozen LLMs/VLMs through learned binary masks and construct token-routed sparse experts for dense-to-MoE upcycling.

Sparse Neural Network Optimization [SA-RST]
- Proposed a sharpness-aware sparse optimization framework that adaptively schedules regularization to improve the trainability of randomly sampled subnetworks without dense pretraining.

Agent-Guided Latent Perception for Embodied Models
- Investigating agent-guided latent perception for embodied models, with a focus on visual representations for embodied decision-making.

Agent Harness Optimization
- Investigating information selection and model–harness co-adaptation to reduce external planning and reflection overhead while preserving long-horizon task performance.

Amazon | **Industrial Robotics Group (IRG)** | *Intern* | North Reading, MA — *May 2026 – Aug. 2026*

***Mentor:** Dr. Kai Gao **Manager:** Dr. Jeroen van Baar*

World Modeling and Robotic Control [UniWALM]
- Developed hierarchical latent planning with subtask-conditioned JEPA-style world prediction and dual-conditioned action generation. Contributed to joint video-action-language diffusion with language-based memory.

Reinforcement Learning for Robot Policy Improvement [AnchorQ]
- Studied sample-efficient robot-policy optimization through flow-reversed action anchors and value-guided refinement, alongside asynchronous agent-guided correction with auxiliary imitation objectives.

Socure | **Computer Vision Team** | *Intern* | Remote — *Jun. 2025 – Aug. 2025*

***Mentor:** Dr. Weqiang Mu **Manager:** Dr. Feng Xiao*

Image Restoration and Robust Visual Decoding
- Developed a reconstruction framework for degraded barcode images, addressing occlusion, blur, and missing visual information to improve decoding robustness and accuracy.

AinnovationLabs | *Applied Scientist* | Remote — *Dec. 2022 – Dec. 2024*

***Manager:** Prof. Yun Raymond Fu*

Acoustic Signal Processing for Healthcare
- Developed acoustic signal-processing methods for sleep-stage identification and sleep-disorder detection (**1 patent**), together with a mobile application for multimodal sleep-data acquisition.

Spatio-Temporal Modeling for 3D Pose Estimation

- Developed a spatio-temporal framework for single-person 3D human pose estimation and constructed a dedicated dataset for motion tracking.

Southeast University | *Research Assistant* | Nanjing, China *Jun. 2019 – Aug. 2019*

Advisor: Prof. Fengqi Si

Multimodal Data Fusion and Forecasting

- Designed a prediction model for atmospheric environmental changes by integrating multimodal environmental data, improving forecasting accuracy and robustness.

SELECTED PUBLICATIONS [Full List]

First-Author

[1] ***Mingyuan Zhang***, Yue Bai, Huan Wang, Yizhou Wang, Qihua Dong, Yitian Zhang, and Yun Fu, "Boosting Large Language Models with Mask Fine-Tuning". [Paper] [Code]

[2] ***Mingyuan Zhang***, Yue Bai, Yifan Wang, Yiyang Huang, and Yun Fu, "Rethinking Fine-Tuning: Unlocking Hidden Capabilities in Vision-Language Models". [Paper]

[3] ***Mingyuan Zhang***, Yue Bai, Zhongruo Wang, Yupin Huang, Yiyang Huang, Hailing Wang, Huimin Zeng and Yun Fu, "MASKerade: Token-Routed Mask Experts for Dense-to-MoE Upcycling". [Paper]

[4] ***Mingyuan Zhang***, Ruining Yang, Yi Xu and Yun Fu, "Sports-Traj++: Unified, Disentangled, and Multi-Modal Trajectory Generation for Multi-Agent Movement in Sports".

[5] ***Mingyuan Zhang***, Jianglin Lu, Yiyang Huang, Yitian Zhang, Yizhou Wang and Yun Fu, "Model Reconfiguration for Adaptation: A Survey of Potential, Limits, and Further Learning".

[6] ***Mingyuan Zhang***, Yiyang Huang, Qihua Dong, and Yun Fu, "From Diagnostic Evidence to Lasting Repair: A Survey of Robot Agents toward Recursive Self-Improvement".

Co-First-Author

* Equal contribution.

[1] Yue Bai*, ***Mingyuan Zhang****, Huan Wang, Zhiqiang Tao, Kunpeng Li, and Yun Fu, "Random Sparse Networks Training with Sharpness-Aware Regularization", ***TPAMI***. [Paper]

[2] Yue Bai*, ***Mingyuan Zhang****, Handong Zhao, and Yun Fu, "Advancing Multi-Modal Foundation Models with Adapter Ensemble Strategies".

ADDITIONAL PROJECTS

- Designed and developed a **mobile experimental logging system** for real-time data collection, enabling researchers to record structured experiment information on mobile devices and streamline the data-collection workflow.
- Designed and developed **Unity-based interactive games**, implementing gameplay mechanics, scene construction, user interaction, and narrative world-building.

ACADEMIC SERVICE

Reviewer:

- Conference: ACMMM 2024, AAAI 2025, UAI 2025, ACMMM 2025, NeurIPS 2025, ICRA 2027
- Journal: TKDD, TPAMI

Teaching:

- TA of EECE5639 Computer Vision, Spring 2026, NEU BOS

SKILLS

Languages: Strong reading, writing, and speaking competencies in English and Mandarin Chinese.

Coding: AI-assisted coding, Python, MATLAB, C.

Web Dev: HTML, CSS.

Tools: Xcode, Blender, Unity, Unreal Engine, LaTeX typesetting and publishing.